Multi-Agent Manufacturing System

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement & Objectives

1.1 Problem Statement

Manufacturing industries rely heavily on processes such as supplier identification, procurement planning, and data analysis. These processes often involve handling large volumes of unstructured and scattered information from multiple sources, which leads to inefficiencies, delays, and human errors.

Traditional systems lack automation and intelligent coordination, requiring manual intervention at each stage—from data collection to report generation. This results in:

- Increased operational cost
- Inconsistent data handling
- Slower decision-making
- Inability to process diverse input modalities (text, image, PDF)

The Agent_Multimodal project addresses these challenges by building a multi-agent system that can ingest and reason over multimodal inputs — text queries, uploaded images, and PDF documents — while automating the full supplier research and report-generation pipeline.

1.2 Project Objectives

- Design a collaborative and intelligent multi-agent multimodal system
- Implement agent specialization for improved efficiency
- Develop a Researcher Agent capable of extracting and validating supplier data from text and web sources
- Develop a Writer Agent capable of structuring data and generating professional reports
- Integrate multimodal input handling: text, images, and PDF uploads via Streamlit
- Establish a standardized hand-off protocol (JSON) for inter-agent communication
- Implement state management and ChromaDB memory for workflow tracking
- Improve accuracy, scalability, and automation in manufacturing processes

1.3 Scope of the Project

Focuses on supplier sourcing, multimodal query processing, and report generation.

In Scope:

- Multimodal data input (text queries, image uploads, PDF documents)
- Data extraction from web and APIs
- Data processing and validation
- Report generation in structured format
- Two primary agents: Researcher Agent and Writer Agent
- ChromaDB for semantic caching and memory
- Streamlit-based user interface

Limitations:

- Real-time industrial hardware integration
- Live ERP system connectivity (planned for future)

2. Proposed Solution

2.1 Key Features

Agent Specialization

– Researcher Agent ® Gathers, validates, and enriches supplier data
– Writer Agent ® Transforms structured JSON into professional reports

Multimodal Input Handling

– Text queries entered via Streamlit chat interface
– Image uploads analysed using GPT-4V / vision-capable LLM
– PDF document uploads parsed and summarised by agents

Intelligent Task Decomposition

– Breaks complex queries into manageable subtasks
– Parallelises independent subtasks where possible

Hand-off Protocol

– Structured JSON format ensures seamless agent-to-agent communication

State Management & Memory

– ChromaDB vector store for semantic query caching
– Tracks task progress and stores intermediate results

Workflow Orchestration

– Ensures proper execution sequence
– Handles dependencies between tasks

Scalability

– Modular design — easily extendable with new agents (e.g., logistics, cost analysis)

2.2 Overall Architecture / Workflow

Architecture Overview

The system is divided into multiple layers:

1. **User Interface Layer (Streamlit)**
   – Accepts text, image, and PDF inputs
   – Displays results in readable, formatted layout
2. **Orchestrator Layer**
   – **Central control unit**
   – **Responsible for task creation, assignment, and agent coordination**
3. **Agent Layer**
   – **Researcher Agent — searches, collects, filters, and validates supplier data**
   – **Writer Agent — formats structured data and generates final reports**
4. **Memory / State Layer (ChromaDB)**
   – **Stores intermediate outputs, task status, and semantic embeddings**
   – **Enables fast retrieval of similar past queries**

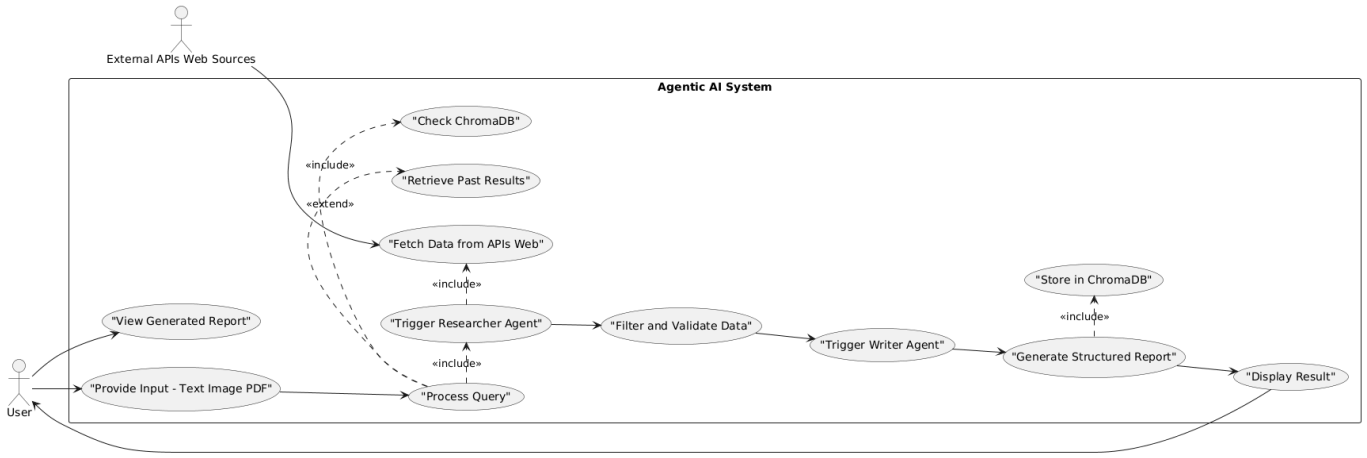
5. **Data Layer**
   - **Web scraping (BeautifulSoup, Selenium)**
   - **REST APIs for live supplier data**
   - **Uploaded PDFs and images processed inline**

Detailed Workflow

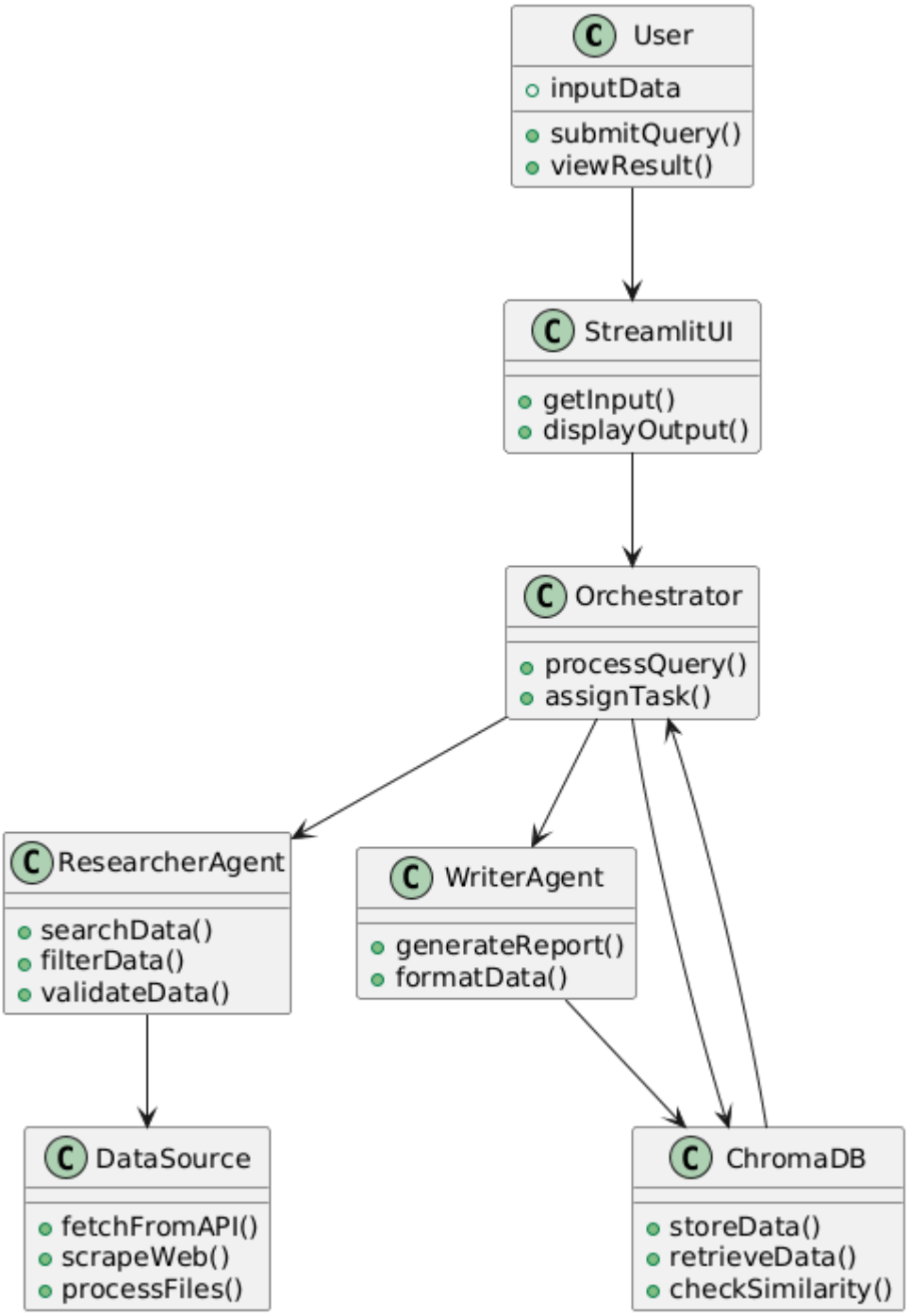
User Input (Streamlit)
↓
ChromaDB Check
(similar query hai?)
↓
Researcher Agent
(GPT-3.5 → JSON)
↓
Writer Agent
(JSON → Report)
↓
ChromaDB Store
(query + result save)
↓
Streamlit Display

UML Diagrams

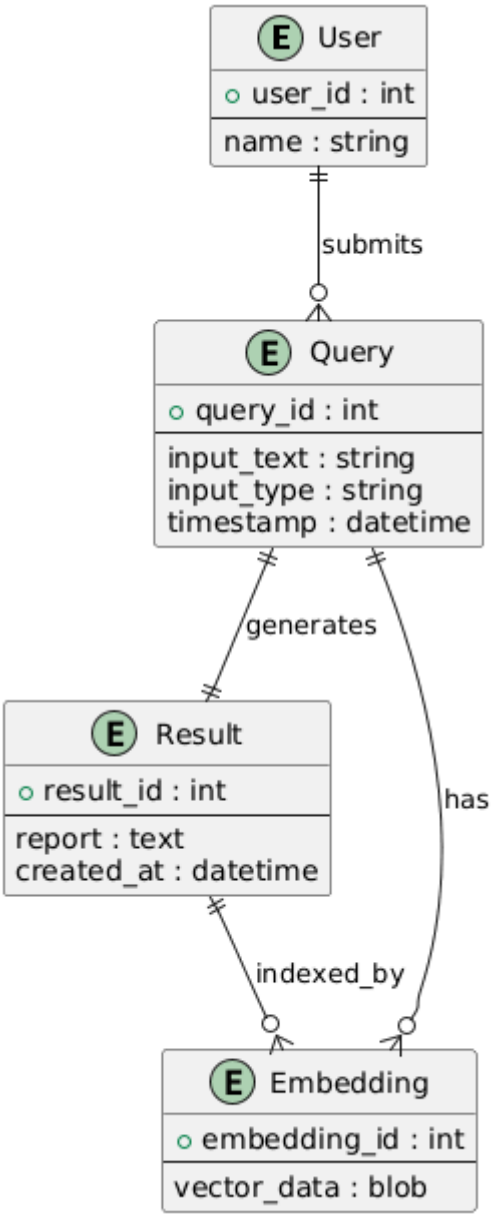
Use Case Diagram



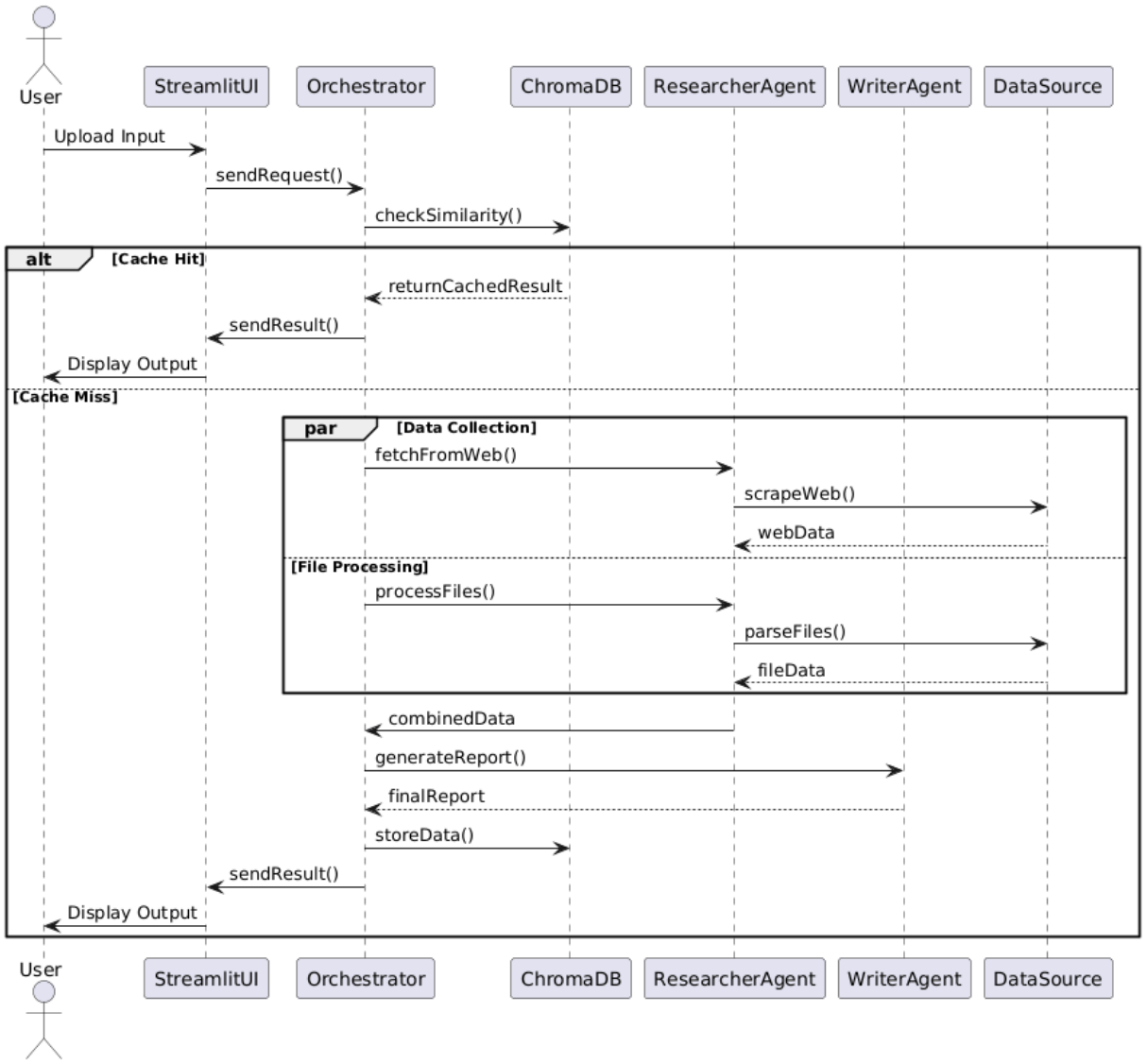
CLASS Diagram



ER Diagram


User
user_id : int
name : string
submits
Query
query_id : int
input_text : string
input_type : string
timestamp : datetime
generates
Result
result_id : int
report : text
created_at : datetime
has
indexed_by
Embedding
embedding_id : int
vector_data : blob

SEQUENCE Diagram



Activity Diagram

2.3 Tools & Technologies Used

Programming & Frameworks

- Python → Core development language
- Streamlit → Interactive multimodal user interface

AI & Agent Frameworks

- LangChain / CrewAI → Multi-agent orchestration
- OpenAI API (GPT-3.5 / GPT-4V) — Text reasoning and vision analysis

Multimodal Processing

- Pillow / OpenCV — Image preprocessing
- PyMuPDF / pdfplumber — PDF text and image extraction
- Base64 encoding — Image payload delivery to vision LLMs

Data Handling

- JSON → Inter-agent data exchange format
- Pandas → Data processing and manipulation

Data Collection

- BeautifulSoup → Web scraping
- Selenium → Dynamic data extraction

Memory & Search

- ChromaDB — Vector database for semantic caching
- FAISS — Alternative efficient similarity search

3. Results & Output

3.1 Screenshots / Outputs

1.Researcher Agent output (raw structured data)

1. List of Suppliers:

1. Hindalco Industries Ltd
   - Location: Mumbai, Maharashtra
   - Price Range: Contact for pricing
   - Lead Time: Varies
2. Vedanta Limited
   - Location: Mumbai, Maharashtra
   - Price Range: Contact for pricing
   - Lead Time: Varies
3. NALCO (National Aluminium Company Limited)
   - Location: Bhubaneswar, Odisha
   - Price Range: Contact for pricing
   - Lead Time: Varies

2.Writer Agent output (formatted report)

2. Pricing Comparison:

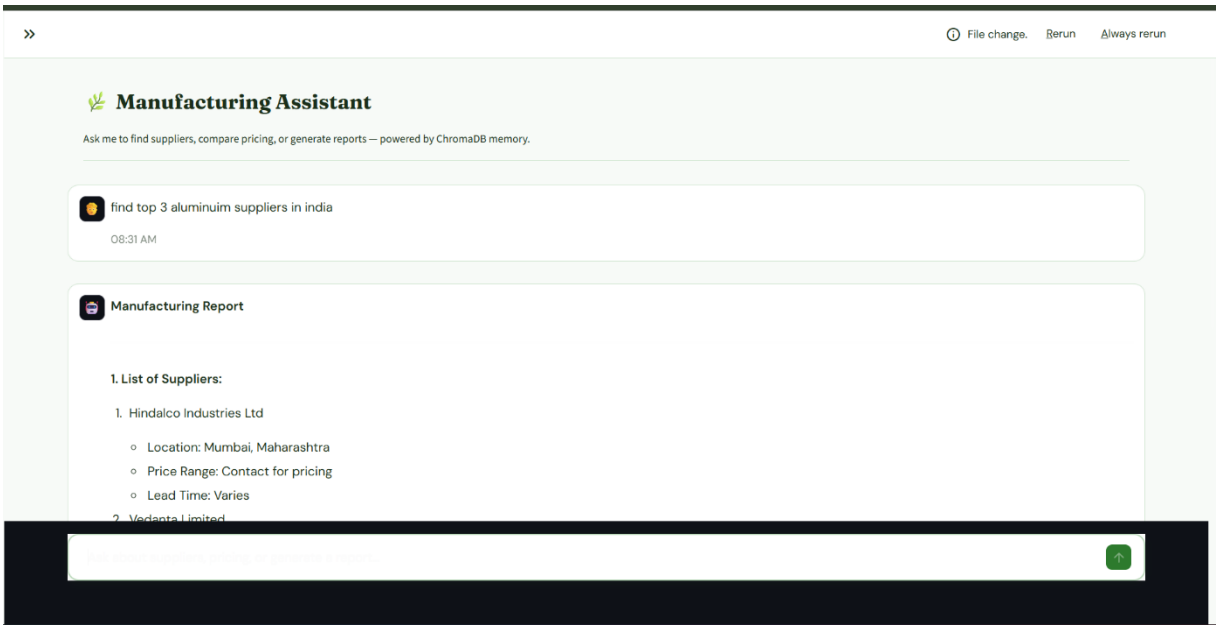
Since all suppliers provide a "Contact for pricing" range and their lead times vary, a direct pricing comparison cannot be made from the available data.

3. Recommendation:

Without specific pricing information and lead times provided in the dataset, it is difficult to recommend the best option. It is recommended to reach out to each supplier to gather detailed pricing, lead time, and any additional services they might offer before making a decision.

This concludes the manufacturing report based on the information provided about the suppliers.

3.Final output dashboard



3.2 Reports / Dashboards / Models

- Supplier comparison reports (tabular format)
- Output fields: Supplier Name · Rating · Location · Price Range · Contact
- Summarised insights: best suppliers, cost-effective options, high-rated vendors
- Dashboard filtering by rating and location
- Simple charts and visualisations (bar, pie)
- Downloadable PDF summary report

3.3 Key Outcomes

- Reduced manual workload significantly through end-to-end automation
- Faster and more accurate data processing via agent specialization
- Demonstrated effective **multimodal agent collaboration** (text + image + PDF inputs)
- Improved **decision-making efficiency** with semantic caching (ChromaDB)
- Achieved **modular and scalable system design** easily extendable to new domains

4. Conclusion

The **Multi-Agent Multimodal Manufacturing System** (Agent_Multimodal) successfully implements a collaborative architecture where specialised agents work together to automate complex manufacturing workflows. By introducing structured JSON communication, semantic memory via ChromaDB, and full multimodal input support — text, images, and PDFs — the system significantly enhances productivity and ensures consistent, high-quality outputs.

This project highlights the potential of multi-agent systems in transforming traditional manufacturing procurement processes into intelligent, automated, and data-driven pipelines. The GitHub repository **github.com/vidhi-1412/Agent_Multimodal** serves as a living codebase demonstrating these capabilities.

5. Future Scope & Enhancements

Additional Specialised Agents

- Cost Analysis Agent
- Inventory Management Agent
- Logistics / Delivery Agent

Enhanced Multimodal Capabilities

- Audio input support for voice queries
- Video analysis for product/supplier demos
- Richer PDF form extraction and table parsing

Live Data Integration

- Real-time supplier APIs for live pricing and availability
- ERP and supply chain management system connectors

Advanced ML Features

- Machine learning models for supplier ranking and recommendation
- Anomaly detection for pricing outliers

Improved UI / UX

- Interactive dashboards with advanced visualization (Plotly, D3.js)
- Role-based access control for enterprise deployment

Domain Expansion

- Logistics
- Healthcare procurement
- Financial services